

INTERNSHIP MONTHLY REPORT TEMPLATE

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Details of the internship schedule: please provide the date. e.g., Week 1 (2/7 - 27/7)

Key date	Key learning activities
Week 1 (2/7 - 7/7)	Getting familiar with PX4 autopilot firmware, I learned about the purpose of the projects. Components and their functions in an UAV
Week 2 (10/7 - 15/7)	Learning to read the flight log, analyze the behavior of the UAV. Learn to manually tune the PID of the quadcopter
Week 3 (18/7 - 22/7)	Install and learn ROS2 through turtlesim (controlling and logging)
Week 4 (22/7 - 27/7)	Merge ROS2 with PX4 autopilot using simulation

1. What skills have you learned in the past 4 weeks? Please describe your progress using the suggestions below:

- I understand the concept of autonomous drone, as well as level of automation and types of UAVs. Main components of a Quadcopter, what's there function, how they contribute to the autonomous flight mission. Also I learned a bit about control (PID) and how that factors effect the drones' behavior. By analysing the log I can understand better what's good and what's need to be fixed.
- Your ability to complete given tasks with timeliness, accuracy, quality and efficiency:
The first week I was given 2 tasked which is investigate a quadcopter to understand it components ans second one was 3D design a battery holder for it. The first task was rather easy because i have already understand quadcopter quite well in my opinion. But the second task cause me some problems : First problems is that i design based on the CAD model of the manufacturer and did not remeasure the real dimension. The quadcopter was modified and my first 3D-printed model was not fit. So I have to extended the finsshing time to second week due to occupied 3D printer. The second version is functionable but still not perfect. Due to the design of the quad leg it's hard to measure correctly by hand, and there is no accurate CAD model, so it was a few centimeters offset downward. In addition I also make a small 3d printed cap for the drone to protect the electronics.
Second and third weeks my task is PID tune the quad, was done in 1 day thanks to the support of mentors. Another task was learning ROS2 which is also not so hard with the support of Chat

GPT and online community. The biggest challenge is that I don't know C++ and have to get the syntax from LLM. I made 2 reports during this task. In the last week of the month I was able to perform SITL, simulate the quadcopter in an ideal environment, and also be able to control it through companion computers and code to send a mission, stream data back to the control station.

- Your teamwork skills and approach method to organizing your work:

In the first month I was doing solo and mostly interacted with my mentors so there is no team yet. However in the 3rd week there is a fourth year student from HUST also enrolled as an intern and he has a similar track to me so I was assigned to instruct him if he has any difficulty because I have some head start.

My approach to work is that analyze the problem, define what I need to solve it, what step it will take, in each step what are the needs, the income and outcome. Try to note everything and keep doing without stopping. If fail then try again, don't stop and if I fail too much then I try another way. If I still fail then I ask for supports.

- Your initiatives and problem-solving skills while completing given tasks

In the battery holder design task at first I was doing some simple solution which involve no mechanical movement but then the supervisor said that it wasn't good enough , the battery can slip in multiple directions, need strapping, etc. So he showed me the version that someone else did before. Based on that I changed my design to something similar, but with different mechanisms, a more simple to design one, and thankfully it works as well, maybe not as good as I intended but still enough.

- In your opinion, what skills do you need to improve to complete the task?

I think I need to analyze the problem better instead of “do first think later” kind of mindset. I think it’s not a good approach but I think it’s also a way to keep me moving forward. That’s maybe because I don't afraid to do it again if I make a mistake.

2. Are there any problems or delays with the internship where you need support?

No, everything is going well and I am doing what I love here.

3. With respect to the final report, with questions on professional responsibilities and on consideration of standards and laws:

- What professional responsibilities have you identified in your last period of internship?

I maintained thorough documentation of all processes, results, and revisions, ensuring that findings were accurately recorded and could be referenced for future work or presentations. I supported the design, testing, and optimization of prototypes, ensuring they met performance requirements and could be further developed into viable products.

- What standards, laws, etc. have you identified in your last period of internship?

Ensuring the safety of both the UAV and its operators was a key priority. I followed general safety protocols during testing, including proper handling of equipment and testing in controlled environments. This helped minimize the risk of accidents and malfunctions.

